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NCE Exam Practice Platform — Mauritius

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Information & Communication Technology

Grade 9 — National Certificate of Education (NCE)

PREDICTED PAPER 4 — Operating Systems, Software, Troubleshooting & Emerging Technologies

Specimen Practice Paper — Not an Official MES Document

Duration: 1 hour 45 minutes | **Total Marks:** 100

Candidate Name: _____

Class / School: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name and class in the spaces provided above.
2. Answer **ALL** questions in all three sections.
3. Section A: Tick **ONE** box only for each multiple-choice question.
4. Section B: Write your answers in the spaces provided.
5. Section C: Use additional paper if the space provided is insufficient.
6. Calculators are **NOT** permitted.
7. Write clearly in black or blue ink; pencil may be used for diagrams only.

Section	Topic	Marks
A	Multiple Choice Questions	25
B	Short Structured Questions	40
C	Extended / Scenario Questions	35
TOTAL		100

SECTION A — Multiple Choice Questions [25 marks]

Tick the ONE correct answer. Each question is worth **1 mark**.

1. The primary role of an operating system is to: 1
- ☐ A run word processing and spreadsheet applications
 - ☐ B manage hardware resources and provide an interface between user and hardware
 - ☐ C connect the computer to the internet
 - ☐ D store application software permanently
2. Which of the following is an example of *system software*? 1
- ☐ A Microsoft Word
 - ☐ B Google Chrome browser
 - ☐ C Device driver for a printer
 - ☐ D A spreadsheet for budgeting
3. *Multitasking* in an operating system means: 1
- ☐ A running multiple CPUs simultaneously
 - ☐ B allowing multiple users to log into the same computer at the same time
 - ☐ C managing multiple processes apparently simultaneously by sharing CPU time
 - ☐ D using multiple monitors with one computer
4. A *Graphical User Interface (GUI)* allows users to interact with a computer using: 1
- ☐ A only typed text commands
 - ☐ B voice commands only
 - ☐ C windows, icons, menus, and a pointer (WIMP)
 - ☐ D punch cards and binary input
5. Open source software is software that: 1
- ☐ A is always free to use and has no licence restrictions
 - ☐ B has its source code made available for anyone to view, modify, and distribute
 - ☐ C can only be used by one person at a time
 - ☐ D is software that has not yet been released to the public

6. Which type of software licence allows a single user on a single machine? 1
- ☐ A Site licence
 - ☐ B Network licence
 - ☐ C Single-user licence
 - ☐ D Freeware licence
7. Virtual memory is used by an operating system to: 1
- ☐ A create a backup of RAM on the internet
 - ☐ B extend available memory by using part of the hard disk as if it were RAM
 - ☐ C run programs without a physical hard disk
 - ☐ D compress files to free up storage space
8. Which of the following would be classified as *application software*? 1
- ☐ A BIOS firmware
 - ☐ B Windows OS kernel
 - ☐ C A photo-editing program
 - ☐ D A network device driver
9. A computer boots slowly and the hard disk light is constantly active. The most likely cause is: 1
- ☐ A The monitor needs replacing
 - ☐ B Too little RAM, causing heavy use of virtual memory (page file)
 - ☐ C The keyboard driver is missing
 - ☐ D The network connection is too slow
10. Which troubleshooting step should be done FIRST when a printer does not print? 1
- ☐ A Replace all ink cartridges
 - ☐ B Check that the printer is powered on and connected
 - ☐ C Reinstall the operating system
 - ☐ D Call a technician immediately
11. The Internet of Things (IoT) refers to: 1
- ☐ A a new type of internet protocol replacing TCP/IP
 - ☐ B everyday physical objects embedded with sensors and connected to the internet
 - ☐ C a network only for industrial machines
 - ☐ D websites that sell physical goods online

12. Cloud computing means: 1
- ☐ A storing weather data on the internet
 - ☐ B delivering computing services (storage, software, processing) over the internet on demand
 - ☐ C using wireless networking within a building
 - ☐ D running all software from a USB drive
13. Which of the following is an advantage of cloud storage over local storage? 1
- ☐ A Always accessible without an internet connection
 - ☐ B No ongoing subscription cost
 - ☐ C Files can be accessed from any device with an internet connection
 - ☐ D Provides faster access than local SSD
14. Machine learning is a branch of computing where systems: 1
- ☐ A follow a fixed set of rules programmed by a developer
 - ☐ B learn from data and improve their performance without being explicitly programmed
 - ☐ C can only process numbers, not text or images
 - ☐ D require a human operator to approve each decision
15. A *utility program* is software designed to: 1
- ☐ A create documents and presentations
 - ☐ B perform specific maintenance tasks such as disk clean-up, antivirus scanning, or file compression
 - ☐ C replace the operating system
 - ☐ D design graphical user interfaces
16. The purpose of *defragmentation* on a traditional HDD is to: 1
- ☐ A remove virus infections from the drive
 - ☐ B rearrange fragmented files so related data is stored contiguously, improving read speed
 - ☐ C permanently delete files from the drive
 - ☐ D encrypt the drive contents
17. Which of the following is a characteristic of *proprietary software*? 1
- ☐ A Source code is freely available to all users
 - ☐ B Users can modify and redistribute it freely
 - ☐ C Owned by a company; users are restricted by a licence agreement
 - ☐ D Always free to download and use

18. A computer's screen suddenly goes blank (black screen) while it is running. The most likely hardware cause is: 1
- ☐ A A software update has completed
 - ☐ B Loose monitor cable or a faulty graphics card
 - ☐ C The keyboard has been disconnected
 - ☐ D The internet connection has dropped
19. Which of the following best describes a *smart device*? 1
- ☐ A Any device that costs more than Rs 5000
 - ☐ B A device with network connectivity and computing capability that can interact with other devices
 - ☐ C A device that can only be controlled by voice
 - ☐ D A mobile phone with a large screen
20. The operating system function that decides which process gets CPU time next is called: 1
- ☐ A Memory management
 - ☐ B File management
 - ☐ C Process scheduling
 - ☐ D Device management
21. Freeware software is: 1
- ☐ A software with source code available to modify
 - ☐ B software available at no cost but with the source code kept private
 - ☐ C software that must be purchased after a trial period
 - ☐ D software bundled with new hardware
22. Robotics technology is increasingly used in manufacturing because robots can: 1
- ☐ A replace all human jobs without any drawbacks
 - ☐ B perform repetitive tasks with high precision and without fatigue
 - ☐ C think creatively and solve novel problems
 - ☐ D communicate emotions to human workers
23. Which utility program is specifically designed to protect a computer from malicious software? 1
- ☐ A Disk defragmenter
 - ☐ B File compression tool
 - ☐ C Antivirus software
 - ☐ D Backup utility

24. A user notices that their Wi-Fi connection keeps dropping every few minutes. After checking that the router is working for other devices, the next best troubleshooting step is to:

1

- ☐ A Replace the entire computer
- ☐ B Update or reinstall the wireless network adapter driver
- ☐ C Cancel the internet subscription
- ☐ D Disable the firewall permanently

25. Augmented reality (AR) differs from virtual reality (VR) because AR:

1

- ☐ A creates a completely immersive computer-generated environment
- ☐ B overlays digital information onto the real-world view seen by the user
- ☐ C requires more powerful hardware than VR
- ☐ D can only be used on desktop computers

SECTION B — Short Structured Questions [40 marks]

Answer **ALL** questions.

B1. Operating System Functions

5

- (a) State *four* functions of an operating system.[4]

- (b) Explain the difference between a *GUI* and a *CLI (Command Line Interface)*. Give one advantage of each. (combined 1 mark for a valid difference stated) [1]

B2. Software Types

5

- (a) Complete the table by classifying each software item. Write **System Software**, **Application Software**, or **Utility Software**. [4]

Software	Type
Windows 11 operating system	
Google Docs (online word processor)	
Antivirus scanner	
Printer device driver	

- (b) Explain the difference between *freeware* and *open source* software.[1]

B3. Troubleshooting

5

- (a) A student's computer displays the message: "*No bootable device found.*" State two possible causes of this error. [2]

- (b) Suggest a logical sequence of troubleshooting steps for a computer that will not connect to a Wi-Fi network. State at least *three* steps in order. [3]

B4. File Management and Utilities

5

- (a) Explain what *file compression* does and give one benefit of compressing files before sending them by email. [2]

- (b) Explain the difference between a *full backup* and an *incremental backup*. [2]

- (c) State one reason why regular backups are important.[1]

B5. Internet of Things (IoT)

5

- (a) Define the term *Internet of Things (IoT)*. [2]

- (b) Give *two* examples of IoT devices used in a smart home and explain what each does. [2]

- (c) State one security risk associated with IoT devices.[1]

B6. Cloud Computing

5

- (a) Describe what is meant by *cloud computing*. [2]

- (b) Give *two* advantages and *one* disadvantage of a school using cloud-based software (e.g. Google Workspace) instead of installing software on each computer. [3]

Advantage 1:

Advantage 2:

Disadvantage:

B7. Emerging Technologies

5

- (a) Explain the difference between *Artificial Intelligence (Ai)* and *machine learning*. [2]

- (b) State one benefit and one ethical concern associated with the widespread use of AI in society. [2]

Benefit:

Ethical concern:

- (c) Explain how *biometric authentication* (e.g. fingerprint scanning) improves security over a traditional password. [1]
-

B8. Legal and Ethical Issues in Computing

5

- (a) The Data Protection Act governs how personal data is collected and used. State *two* rights it gives to individuals regarding their personal data. [2]
-

- (b) A student downloads a copyrighted song from the internet without paying. State which law this may break and explain why. [2]
-

- (c) Define *cyberbullying* and state one way schools can help prevent it.[1]
-

SECTION C — Extended / Scenario Questions [35 marks]

C1. Scenario: Upgrading a School's Computer Suite

12

The ICT department at Bright Future Secondary School is upgrading its computer suite of 30 workstations. Currently, each computer runs an outdated version of a proprietary operating system and all software is installed locally. The ICT manager is considering switching to **cloud-based software** and **open source operating systems**. Each computer currently has 4 GB RAM, a 5400 rpm HDD, and a dual-core processor.

- (a) Explain the difference between *proprietary* and *open source* operating systems. Give one example of each. [4]

- (b) Discuss **two advantages** and **two disadvantages** of switching from locally installed software to cloud-based software for the school's 30 computers. [4]

Advantages	Disadvantages
1.	1.
2.	2.

- (c) The ICT manager decides to upgrade the RAM from 4 GB to 16 GB and replace the HDDs with SSDs. Explain how each upgrade will improve the user experience for students using the computers. [4]

RAM upgrade (4 GB → 16 GB):

HDD → SSD upgrade:

C2. Scenario: Smart Home System — IoT and Emerging Technology

12

The Ramkisoorn family has installed a smart home system including: a smart thermostat that learns their temperature preferences; security cameras with motion detection that alert their phones; smart locks with fingerprint access; and a voice-controlled assistant that controls lighting and appliances. All devices are connected via their home Wi-Fi and managed through a single smartphone app.

- (a) For each smart device listed, identify what technology it uses and explain one benefit it provides. Complete the table. [4]

Device	Technology used	Benefit
Smart thermostat		
Security camera		
Smart lock		
Voice assistant		

- (b) Identify **three cybersecurity risks** associated with this smart home system. For each risk, suggest one way to reduce it. [6]

Risk	How to reduce it
Risk 1:	
Risk 2:	
Risk 3:	

- (c) The smart thermostat "learns" the family's preferences. Explain how *machine learning* makes this possible. [2]

C3. Troubleshooting and System Performance

11

A technician is called in to assess the following problems at a small office:

Problem A: One computer shows a blue screen of death (BSOD) every time a large file is saved to the HDD.

Problem B: The internet works on all computers except one. That computer shows "Limited connectivity" and cannot load any web pages.

Problem C: All computers in the office are running slowly. Opening any application takes over 2 minutes.

- (a) For **Problem A**: suggest the most likely cause and describe two steps the technician should take. [3]

(b) For **Problem B**: list *four* diagnostic steps in logical order to identify and fix the connectivity issue on that one computer. [4]

(c) For **Problem C**: suggest *two* possible causes for the slowness of all computers in the office, and recommend one software-based solution for each. [4]

Cause 1 + Solution:

Cause 2 + Solution:

MARK SCHEME — Paper 4

Accept any equivalent correct answer. Troubleshooting steps may vary — award marks where the response demonstrates logical problem-solving methodology. For legal/ethical questions accept relevant examples from the Mauritian context.

SECTION A — Multiple Choice

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	C	3	C	4	C	5	B
6	C	7	B	8	C	9	B	10	B
11	B	12	B	13	C	14	B	15	B
16	B	17	C	18	B	19	B	20	C
21	B	22	B	23	C	24	B	25	B

SECTION B

Q	Expected Answer	Marks
B1(a)	Any 4 of: process management/scheduling; memory management; file system management; device management (drivers); providing a user interface; security/access control [1 each]	4
B1(b)	GUI uses graphical elements (windows, icons, mouse) — easier for beginners; CLI uses typed commands — faster for experts / more powerful for scripting (any valid contrast)	1
B2(a)	Windows 11: System Software; Google Docs: Application Software; Antivirus scanner: Utility Software; Printer driver: System Software [1 per correct pair]	4
B2(b)	Freeware: free to use but source code is closed / cannot be modified; open source: source code is available and can be freely modified and redistributed	1
B3(a)	Any 2 of: corrupted or missing OS boot files; hard disk failure; boot order set to wrong device in BIOS; HDD not connected properly [1 each]	2
B3(b)	Step 1: Check Wi-Fi is enabled on the device / toggle off then on [1]; Step 2: Forget the network and reconnect / re-enter password [1]; Step 3: Restart the router or check if other devices connect / update network driver [1]	3
B4(a)	File compression reduces the size of a file using algorithms that remove redundancy [1]; smaller file transfers faster / uses less email attachment quota [1]	2
B4(b)	Full backup: copies all files every time [1]; incremental backup: only copies files that have changed since the last backup — faster and uses less storage [1]	2
B4(c)	To restore data after hardware failure / ransomware / accidental deletion / theft (any one valid reason)	1

B5(a)	IoT is a network of physical devices embedded with sensors, software, and connectivity [1] that enable them to collect and exchange data with other devices and systems via the internet [1]	2
B5(b)	Smart thermostat: senses temperature and adjusts heating/cooling remotely [1]; smart bulb: can be turned on/off/dimmed via app; smart doorbell: streams live video to phone; smart plug: remotely controls power to appliances (any 2 valid IoT devices with function [1 each])	2
B5(c)	Weak default passwords can allow unauthorised access; unencrypted data can be intercepted; poorly updated firmware can be exploited; one compromised device can give access to the whole network (any one)	1
B6(a)	Cloud computing delivers computing services (storage, software, servers, processing) over the internet [1] on demand, without the user needing to own or manage the underlying infrastructure [1]	2
B6(b)	Advantages (any 2): always up to date; accessible from any device; lower upfront cost; automatic backup; easy collaboration [1 each]; Disadvantage: requires internet connection to function; data privacy/security concerns; ongoing subscription cost (any 1) [1]	3
B7(a)	Artificial intelligence is the broad field of making machines simulate human intelligence [1]; machine learning is a subset where systems learn from data and improve without explicit programming [1]	2
B7(b)	Benefit: faster medical diagnosis / improved accessibility / efficiency gains / personalised services (any one) [1]; Ethical concern: job displacement / bias in decision-making / privacy invasion / lack of accountability (any one) [1]	2
B7(c)	A biometric feature (e.g. fingerprint) is unique to the individual and cannot be guessed or stolen as easily as a password / cannot be shared	1
B8(a)	Any 2 of: right to access data held about them; right to have inaccurate data corrected; right to have data deleted; right to know how data is used [1 each]	2
B8(b)	Copyright law (or similar — accept Computer Misuse Act if appropriately explained) [1]; the song is protected intellectual property and downloading without permission/payment violates the creator's rights [1]	2
B8(c)	Cyberbullying: using digital technology to harass, threaten, or humiliate someone [0.5]; School measure: acceptable use policy / reporting systems / digital literacy education / monitoring of online platforms [0.5]	1

SECTION C

Q	Expected Answer / Mark Points	Marks
C1(a)	Proprietary: owned by a company; source code is closed; users pay a licence e.g. Microsoft Windows [2]; Open source: source code is public; free to modify and distribute e.g. Linux/Ubuntu [2]	4

C1(b)	Advantages (any 2 from): no installation needed; always up to date; accessible from anywhere; collaboration in real-time; reduced IT maintenance burden [1 each]; Disadvantages (any 2 from): requires reliable internet; data privacy concerns; subscription costs; limited offline functionality [1 each]	4
C1(c)	RAM: more RAM means more applications can be held in memory simultaneously [1]; less swapping to virtual memory on disk — programs open and respond faster [1]; SSD: much faster read/write speeds than HDD [1]; OS and programs load significantly faster; boot time reduced dramatically [1]	4
C2(a)	Thermostat: machine learning / sensors [1] — automatically adjusts temperature saving energy [1]; Camera: motion sensing / image processing [1] — real-time alerts improve home security [1]; Smart lock: biometric / fingerprint sensor [1] — keyless entry with personalised access [1]; Voice assistant: natural language processing / voice recognition [1] — hands-free control of home devices [1] (1 per technology + 1 per benefit, max 4 marks total for table — 1 per row)	4
C2(b)	Risk 1: Weak/default passwords — change default passwords, use strong unique credentials [2]; Risk 2: Unencrypted data transmission — enable WPA3 Wi-Fi encryption / use VPN [2]; Risk 3: Outdated firmware — enable automatic firmware updates on all devices [2] (1 per risk + 1 per countermeasure)	6
C2(c)	The thermostat collects data on when and how the family adjusts temperature [1]; machine learning algorithms analyse these patterns and build a model that predicts and automatically sets preferred temperatures without being explicitly programmed [1]	2
C3(a)	Cause: failing or corrupt HDD / bad sectors on disk [1]; Step 1: Run a disk diagnostic tool (e.g. chkdsk / SMART test) [1]; Step 2: Back up data immediately and replace the HDD if errors are found [1]	3
C3(b)	Step 1: Check physical connection (Ethernet/Wi-Fi adapter enabled) [1]; Step 2: Run network troubleshooter / release and renew IP address (ipconfig /release /renew) [1]; Step 3: Check DNS settings / try setting to 8.8.8.8 [1]; Step 4: Update or reinstall network adapter driver [1]	4
C3(c)	Cause 1: Malware running in background — Solution: run full antivirus scan and remove threats [2]; Cause 2: Too many startup programs / fragmented HDDs — Solution: disable unnecessary startup programs / run defragmentation utility [2] (1 per cause + 1 per solution)	4

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